

Visual Deception Detection for Financial Security: Homoglyph and Deepfake Identification

by

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ABSTRACT

Visually-deceptive text, manifesting through homoglyph attacks, poses a growing threat across cybersecurity, financial systems, and identity verification. While existing string detection methods are easily scalable, they fail to capture the nuanced perceptual similarities between textual inputs. This paper introduces Visually-Aligned Text Embeddings (VA-TE), a contrastive learning framework that uses curriculum learning to train a lightweight projection head on top of a pretrained vision-language model (VLM) text encoder, translating semantic embeddings into representations aligned with human visual perception. Crucially, this method operates directly on text, enabling faster, lighter-weight detection than prior image-based approaches that require rendering and training on millions of images. Evaluated on a large-scale homoglyph benchmark, VA-TE attains competitive performance ($AUC = 0.95$) using a SigLIP encoder backbone, approaching state-of-the-art (SOTA) image-based methods while offering substantial gains in scalability and deployment efficiency. Further, fusing these visually-aligned embeddings with complementary string-similarity features yields SOTA results ($AUC = 0.98$), underscoring the value of multi-modal signals for robust spoof detection. Taken together, VA-TE establishes a self-supervised procedure for converting semantic text features into visually grounded representations applicable across perceptual string-matching tasks.

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Chapter 1

Introduction

1.1 Motivation

The proliferation of digital content has brought with it an escalating challenge: visual deception. This threat manifests critically in financial systems, where adversaries exploit visual similarities to bypass security measures. Two problems exemplify this challenge. First, in homoglyph attacks, adversaries create visually similar account names to deceive users into trusting fraudulent accounts. These spoofed names look nearly identical to legitimate companies, fooling human observers, while evading detection systems that rely on text-based string matching rather than visual appearance. Second, AI-generated synthetic media, including deepfakes, threaten biometric verification and identity authentication systems that rely on facial recognition. These deception techniques span multiple areas and types of fraud, yet share a common strategy: exploiting the gap between visual appearance and underlying representation to deceive both humans and automated systems. Given the scale and reach of these threats, robust and accurate detection systems are essential to maintain the integrity of modern financial infrastructure, identity verification platforms, and information ecosystems that depend on content authenticity. Traditional detection methods have approached these challenges by identifying specific artifacts or patterns characteristic of deceptive content. In the text domain, string matching algorithms rely on character-level metrics such as edit distance and token overlap, treating homoglyphs as a text problem when the deception operates visually. In the audio-video domain, deepfake detectors learn to recognize generation artifacts left by specific models. However, this artifact-driven paradigm suffers from a critical limitation: adversaries continuously evolve their techniques, rendering artifact-based detectors brittle. A detector trained to recognize artifacts from one generative model often fails catastrophically when confronted with outputs from newer, unseen models. As generative capabilities advance, the cat-and-mouse game of artifact detection becomes increasingly unsustainable. This thesis proposes a paradigm shift: rather than chasing the ever-changing artifacts of deceptive content, we instead learn the stable, intrinsic properties that characterize authentic content. This reframing transforms detection from a reactive pursuit of adversarial signatures to a proactive learning of what genuine content looks like. By focusing on the commonalities shared by real images or the natural alignment between text semantics and visual appearance, we develop detectors that generalize across deception

methods.

1.2 Thesis Overview

This thesis addresses visual deception for two distinct tasks, developing specialized approaches for each:

1.2.1 Part I: Textual Visual Deception (Chapter 2)

We tackle homoglyph attacks, where adversaries substitute visually similar characters to evade text-based spoof detection. A naive visual solution might treat text as images, rendering strings and applying computer vision models, but this proves impractical at scale due to the computational overhead and memory requirements of image processing for every account registration or transaction. Instead, we introduce Visually-Aligned Text Embeddings (VA-TE), which exploits a key insight: Vision-Language Models (VLMs) pretrained on image-text pairs learn text encoders that implicitly capture visual characteristics of text appearance. By fine-tuning the VLM text encoder directly, VA-TE encodes visual properties from text itself, without rendering images, enabling fast, memory-efficient inference at deployment scale. Through contrastive learning with curriculum-based hard negative mining, VA-TE learns to embed visually similar strings close together in representation space, even when they are distant according to character-based string metrics. Combined with complementary string-based features in an ensemble, VA-TE achieves 98% average precision, matching or exceeding vision-based methods while offering substantial advantages in speed, memory footprint, and deployability for real-world systems.

1.2.2 Part II: Image-Level Visual Deception (Chapter 3)

We address deepfake detection in the era of rapidly evolving generative models. Existing detectors overfit to artifacts specific to training-set generative models and fail on unseen model families. We propose Intrinsic Authenticity Detection, which learns the properties shared exclusively by real audio-visual content through a self-supervised disentanglement framework. Specifically, IAD separates homogeneous features (common across all real samples) from heterogeneous features (sample-specific content) isolating authenticity signals from content-related information in the input representations. A temporal classifier then uses these homogeneous authenticity features to detect fakes by assessing proximity to real sample properties rather than memorizing artifacts of specific generative models. Evaluated on a comprehensive dataset spanning GANs, diffusion models, and multi-step synthesis methods, NTF achieves **XXX** % mean average precision with superior cross-model generalization compared to state-of-the-art baselines.

While these two problems differ substantially in their technical domains, from text embeddings to image forensics, they share a unifying methodological philosophy: learning stable representations of authentic content, enabling more robust detection systems.

1.3 Contributions

Chapter 2

Visually-Aligned Text Embeddings for Homoglyph Detection

2.1 Introduction

Research exploring visual similarity of text has remained mostly untouched, despite vast applications within cybersecurity [1], website infrastructure [2,3], bank check processing [4], and other growing digital fields. Financial institutions must develop systems protecting against scammers, who subtly manipulate legitimate company names to create visually-similar “spoofs” that trick customers and merchants into approving fraudulent transactions. Businesses centered around user accounts must create software designed to prevent username impersonations [5], while financial technology companies require systems to ensure precise digital check processing. Each of these cases demands high accuracy, minimizing false negatives that prevent malicious visually-similar text from scraping through, while avoiding false positives that cause significant customer inconvenience.

A common spoofing tactic is the use of *homoglyphs*, characters from different writing systems that look identical but have distinct Unicode code points. For instance, the Latin letter “a” (U+0061) and the Cyrillic “a” (U+0430) appear the same to readers but not to computers. Similarly, attackers may substitute the letter “O” with the digit “0,” or replace “e” with accented forms such as “é.” Such substitutions exploit human perception while bypassing naive string matching, making homoglyph detection essential for spoof prevention.

Despite this problem’s relevance, existing text similarity research focuses on semantic relationships [6–9] —aligning words and phrases based on meaning rather than visual appearance. Methods for detecting visually-similar text have lagged behind industry needs, with most companies using string matching techniques to fulfill various mission-critical detection tasks. These methods include string similarity metrics like Levenshtein distance [10], which define decision boundaries based on the number of edits required to match two names. Fuzzy matching techniques are also used to assess relative similarity, where names with high similarity scores are flagged as potentially fraudulent. Typically, newly registered text, including account registrations and usernames, is iteratively checked against a reference dataset, with higher similarity indicating visually-similar text.

String-based similarity detection methods face a major challenge: they fail to capture

the perceptual nuances of string modifications, as these are intentionally designed to be visually subtle to evade downstream detection. Methods that are currently being used can detect minor textual alterations, but often create false positives when string lengths become smaller [11]. This necessitates the development of more sophisticated methods that align with human-like perception of visual string similarity, specifically a scalable and deployable implementation capable of training accurate models without using large quantities of image-based data. In practice, attackers rarely rely on a single deception strategy, blending visual similarity with textual manipulation patterns. A robust detection system must therefore capture both dimensions of deception—perceptual similarity and structural variation—to defend against a spectrum of spoofing attacks.

We propose Visually-Aligned Text Embeddings (VA-TE), a contrastive learning [12] framework that expands upon pre-trained vision-language models [13] capable of encoding the visual characteristics of text. Through building on models that generate semantically rich embeddings directly from text, this approach captures subtle, visually deceptive name variations, achieving significantly higher separability than string-based methods, improving ROC-AUC from 0.81 [14] to 0.95, while also offering deployment gains compared to existing machine learning approaches [14,15]. To fulfill this task, we fine-tune a lightweight projector [16] on top of these pre-trained embeddings to enhance the system’s ability to distinguish between legitimate names and visually-similar spoofs.

We use contrastive learning [12], experimenting with various contrastive objectives [17–19] to optimize similarity in our target embedding space. Additionally, we employ curriculum learning [20] to progressively introduce more challenging negative examples during training, enabling the model to learn nuanced decision boundaries that improve sensitivity to string variations. To evaluate the effectiveness of VA-TE, we frame spoof detection as our primary evaluation task, measuring the quality of our learned embeddings through their ability to distinguish between legitimate company names and visually deceptive spoof counterparts.

This work presents a novel approach to visually-aligned text embeddings, marking an important step toward multi-modal representations that integrate the visual characteristics of text, reflecting how humans read and interpret it. Building on recent advances in embedding refinement [21], our contribution introduces a visually grounded text representation that captures perceptual detail, enabling state-of-the-art performance in visual text similarity tasks.

The paper is structured as follows: Section ?? reviews relevant literature, spanning traditional string similarity methods, contrastive learning techniques, and vision-language models. Section ?? details the proposed method, including architectural design and training objectives. Section ?? describes the datasets, evaluation protocols, and baseline comparisons, while Section ?? presents experimental results. Section ?? discusses avenues for future work, and Section ?? concludes with a summary of our findings.

2.2 Related Works

2.2.1 String Matching Methods

Despite challenges with accuracy and implementation [11], string-based similarity methods act as fast, computationally efficient proxies for evaluating visual similarity between strings, demonstrating their value as components in similarity detection systems [22]. These methods fall into two main categories: edit distance metrics and fuzzy matching methods, each capturing different aspects of textual manipulation patterns commonly exploited in spoofing attacks. While individually limited, their distinct strengths and failure modes make them well-suited for ensemble-based approaches.

Levenshtein Distance Metrics

Levenshtein distance [10] measures the minimum number of single-character edits (insertions, deletions, and substitutions) required to transform one string into another. This metric excels at detecting minor manipulations in longer strings. However, edit distance exhibits systematic failures that limit its standalone effectiveness. For short strings, legitimate variations and unrelated names can trigger false positives due to the metric’s insensitivity to string length. A single character difference between genuinely distinct entities can produce a low edit distance that incorrectly suggests spoofing. Conversely, sophisticated spoofing attacks on longer strings can involve many character edits while maintaining visual similarity, resulting in high edit distances that evade detection. This can be seen in Table 2.1.

Table 2.1: Levenshtein distance misclassifications (threshold of ≤ 2 edits).

Levenshtein Distance Failure #1	Levenshtein Distance Failure #2
‘3M’ vs ‘H&M’	‘Microsoft’ vs ‘Microsofts.’
Distance = 2 $\rightarrow$ Spoof (Wrong)	Distance = 3 $\rightarrow$ Non-Spoof (Wrong)

Despite these limitations, Levenshtein distance achieves reasonable performance as a standalone classifier, with reported ROC-AUC of 0.81 on domain name spoofing tasks [14].

Fuzzy Matching Methods

Percentage-based similarity metrics, including Token Sort Ratio, Partial Ratio, and Token Set Ratio¹, calculate similarity based on token overlap and string length normalization. These methods address edit distance’s length sensitivity by producing normalized similarity scores, making them particularly effective for longer strings where proportional similarity is more meaningful than absolute character differences.

¹Fuzzy Matching library, including Token Set Ratio, can be found here: <https://rapidfuzz.github.io/RapidFuzz/Usage/fuzz.html>

While commonly used by financial institutions, fuzzy matching methods exhibit complementary failure patterns that limit their standalone effectiveness. When strings share common words or substantial token overlap, percentage metrics yield high similarity scores that create false positives for genuinely distinct entities. Conversely, well-designed spoofs that appear visually similar but are textually dissimilar, particularly in short strings, often evade detection entirely, resulting in false negatives. Examples are shown below in Table 2.2.

Table 2.2: Token set ratio misclassifications (threshold of ≥ 80).

Token Set Ratio Failure #1	Token Set Ratio Failure #2
‘Invest Co’ vs ‘Square Invest Co’	‘Capital One’ vs ‘Capito1 One’
Ratio = 100 $\rightarrow$ Spoof (Wrong)	Ratio = 55 $\rightarrow$ Non-Spoof (Wrong)

Fuzzy matching achieves superior standalone performance compared to editing distance, with a reported ROC-AUC of 0.86 on domain spoofing tasks [14], reflecting its improved handling of string length variations.

Complementary Strengths and Ensemble Potential

An analysis of failure conditions reveals that edit distance and fuzzy matching offer complementary strengths. Levenshtein distance performs well on short strings with character-level manipulations, where fuzzy matching often fails. Conversely, for longer strings with proportionally similar tokens, fuzzy matching excels while edit distance struggles. These patterns suggest that combining both metrics in an ensemble approach could capture a wider range of spoofing behaviors than either method alone.

Despite these strengths, both techniques fundamentally fail to account for the perceptual subtleties of visual similarity, an essential aspect of effective spoofing. This limitation motivates the development of more advanced methods that embed visual characteristics into their representations to improve similarity analysis and spoof detection.

2.2.2 A Visual Non-ML Approach: HitZone Mapping

HitZone Mapping [23] assesses visual similarity by dividing characters into grids and measuring the degree of overlap between corresponding squares. Each character is mapped onto a grid with varying degrees of granularity, allowing for detection of minor visual differences and providing an interpretable framework for spoof detection. While HitZone maps have been successfully applied at the character level, they have not been widely explored in the context of word-level similarity detection.

When comparing different length strings, words must be carefully aligned so hitboxes match-up on a character-to-character basis. This task would require a dynamic programming approach that hasn’t yet been formally developed.

2.2.3 Embeddings for Similarity Measurement

Unlike the previously mentioned methods, embeddings provide an efficient and continuous framework for similarity assessment, most commonly cosine similarity [6,24]. Learning a task-specific embedding space enables effective thresholding and ranking, both of which are essential for real-time retrieval and large-scale systems [25].

Accordingly, embeddings have been used for similarity assessment across many different tasks. Within computer vision, they power systems for facial recognition [17] and image classification [26], both of which require precise ranking of visually-similar features and items. Extending even to symbols, embeddings have enabled tasks like sign language recognition [27] and handwritten text retrieval [28,29], where visual patterns rather than strict symbolic sequences define similarity. In text-based tasks, embeddings have achieved widespread usage in word semantic settings that measure meaning-based closeness [6,8]. The overwhelming literature outlining the success of embeddings in similarity analysis [30] directly motivates the usage of embeddings for visual text similarity analysis.

2.2.4 Contrastive Learning

Contrastive learning [12] has been effectively used to learn task-specific embeddings requiring fine-grained distinction[31], including face forgery detection[32], text-image alignment[7], and self-supervised representation learning[16].

In particular, SimCLR [16] presents a simple yet powerful framework for contrastive learning of visual representations, using data augmentation to generate positive pairs, a neural network encoder to compute embeddings, and a projection head to map these embeddings into a space where contrastive loss is applied. This framework directly motivates the methodological approach presented in this paper.

Additionally, prior work has applied contrastive learning frameworks, specifically using Siamese convolutional neural networks [33], to the problem of homoglyph detection by learning visually-aligned representations of text strings [14,15]. These studies demonstrate the ability of contrastive loss for distinguishing visually similar strings, underscoring the relevance of contrastive learning for spoof detection tasks.

In contrast to classification-based training, which learns embeddings incidentally while optimizing for label prediction [34], contrastive learning explicitly shapes the embedding space to align with meaningful notions of similarity by bringing similar (positive) samples closer and pushing dissimilar (negative) samples apart [12,35]. The goal is to learn a feature space in which similar inputs are mapped to nearby points, while dissimilar inputs are separated by a significant margin, using distance metrics like cosine distance.

A range of loss functions have been developed to implement this learning objective:

- **Contrastive Loss:** [12] This formulation penalizes positive pairs that are far apart and negative pairs that are too close, typically including a margin hyperparameter to encourage strong separation without collapsing the embedding space.

$$\mathcal{L}_{\text{contrastive}} = y \|z_i - z_j\|^2 + (1 - y) [m - \|z_i - z_j\|]_+^2 \quad (2.1)$$

where $y = 1$ for positive pairs and $y = 0$ for negative pairs, and m is the margin.

- **Triplet Loss:** [17] Extends the contrastive formulation to anchor-positive-negative triplets, ensuring the anchor is closer to the positive than the negative by a margin.

$$\mathcal{L}_{\text{triplet}} = \max(0, \|\mathbf{z}_a - \mathbf{z}_p\|^2 - \|\mathbf{z}_a - \mathbf{z}_n\|^2 + \alpha) \quad (2.2)$$

where $\mathbf{z}_a$, $\mathbf{z}_p$, and $\mathbf{z}_n$ are the anchor, positive, and negative embeddings, and α is the margin.

- **InfoNCE:** [18] Commonly used in self-supervised learning frameworks, InfoNCE generalizes the contrastive objective to a multi-negative classification task, maximizing the similarity of the anchor-positive pair relative to a set of negatives and encouraging discriminative representation learning in large batches.

$$\mathcal{L}_{\text{InfoNCE}} = -\log \left(\frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a \in A(i)} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)} \right) \quad (2.3)$$

where τ is the temperature and $A(i)$ includes one positive and multiple negatives.

- **Supervised Contrastive Loss (SupCon):** [19] Extends InfoNCE to leverage batches of positive samples. This is particularly beneficial in spoof detection where multiple spoof variants may exist for a single legitimate entity.

$$\mathcal{L}_{\text{sup}} = -\sum_{i \in I} \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i^\top z_p / \tau)}{\sum_{a \in A(i)} \exp(z_i^\top z_a / \tau)} \quad (2.4)$$

where $P(i)$ is the set of positives for anchor i , excluding itself.

A critical component of effective contrastive learning is the use of hard negatives [35,36] - negative samples that are close to positives in the initial embedding space before applying contrastive learning. By challenging the model to distinguish between such near-positives, hard negatives refine decision boundaries and encourage the learning of robust, discriminative representations. This consideration naturally motivates the use of curriculum learning strategies, described in the following section.

2.2.5 Curriculum Learning

Curriculum learning [20] is a training paradigm that structures the learning process by gradually increasing the difficulty of training examples over time, mimicking the way humans learn progressively from simpler to more complex tasks. Originally introduced in natural language processing and computer vision tasks, curriculum learning has since been applied to deep metric learning [37] and contrastive representation learning [38], where it has been shown to improve convergence, generalization, and robustness of the learned embedding spaces.

In contrastive learning specifically, where models are trained to bring positive pairs closer and push negative pairs apart in the embedding space, the selection and ordering of training

examples, particularly negative samples, plays a critical role in shaping model performance. A naïve strategy that samples negatives uniformly at random may lead to uninformative or overly trivial contrastive signals [Wang2022], which in turn limit the model’s ability to develop nuanced decision boundaries. This is especially problematic in tasks like homoglyph detection, where many adversarial examples are intentionally designed to be visually deceptive, requiring the model to capture subtle typographic cues.

To address this, recent studies in curriculum-based contrastive learning propose incorporating progressively harder negative samples over the course of training [36]. This structured training helps the model refine its embedding space to capture more fine distinctions. This concept is closely related to hard negative mining [17,35], a well-established technique in deep metric learning, where negatives that are difficult to distinguish from positives are prioritized to accelerate learning and improve discriminative capacity.

Several curriculum learning strategies have been explored in prior work:

- **Curriculum Learning** [20] introduces samples incrementally based on a difficulty metric (e.g., loss, similarity score), allowing the model to focus first on easier instances and later adapt to harder ones.
- **Self-Paced Curriculum Learning (SPL)** [39] ranks training samples based on their contribution to model loss, ensuring continued focus on examples where the model struggles most.
- **Bandit-Based Adaptive Sampling** [40] applies multi-armed bandit frameworks to dynamically allocate sampling effort across difficulty tiers, selecting the most informative samples based on learning progress.

These strategies have demonstrated improvements across various domains, including image retrieval [41], face verification [37], and few-shot classification [42]. However, their use in vision-language pretraining and visually-aligned text representation learning remains unexplored. In our work, we build on these principles and extend curriculum learning into the realm of textual visual-similarity analysis by integrating it with contrastive learning over pre-trained embeddings.

By leveraging curriculum learning to shape the training trajectory of hard negative examples, we repurpose a pretrained text encoder—originally optimized for semantic alignment—to instead encode visual similarity between text strings.

2.2.6 Vision-Language Models

Vision-Language Models (VLMs) [13] have recently emerged as powerful tools for learning joint representations of images and text, finding applicability for image-text matching, image-to-text generation, and zero-shot classification [43]. Trained on large datasets, typically of image-caption pairs [7], these models align semantic information in a shared embedding space, making them effective at recognizing meaning across both modalities. While most VLMs are trained for semantic alignment, prior work has shown that visual features of text can emerge in their internal representations [44], and that semantic similarity in text often correlates with visual similarity [45]. This motivates our investigation into whether such models, when

enhanced with projection layers [16], can be effectively adapted for VA-TE. The following examples illustrate cases where semantic similarity aligns with visual similarity (Table 2.3).

Table 2.3: Examples of strong visual-semantic alignment.

Word 1	Word 2	Visual-Semantic Relationship Explanation
Doggy	Doggie	Visually similar spellings with identical semantic meaning
Google	Go0gle	Visually similar company names with high semantic overlap

Alongside visual-semantic correlation, many VLMs are trained on images that contain naturally occurring text [7], like street signs, logos, and item labels, that are paired with text that describe them. This indirect exposure to textual appearance in visual contexts allows the model to learn association between word meaning and visual appearance [44]. These two factors motivate our usage of VLMs for detecting *perceptual* similarity of text.

To determine our model-specific approach, we categorize VLMs into four architectural types: contrastive-based, VLMs with masking objectives, generative-based, and VLMs from pretrained backbones[46].

- **Contrastive-based VLMs** (CLIP, OpenCLIP, SigLip, ALIGN) use separate encoders for image and text modalities, aligning their outputs through a contrastive loss function.
- **VLMs with masking objectives** (FLAVA) process image and text together through a single transformer with early fusion and cross-modal attention, making them especially effective for tasks that require understanding detailed interactions between different modalities.
- **Generative-based VLMs** (BLIP, BLIP-2, CoCa) combine a vision encoder with a generative text decoder. CoCa, in particular, fuses together dual-encoder retrieval with captioning capabilities.
- **VLMs from Pretrained Backbones** (LLaVA and GPT-4V) integrate visual inputs through adaptors or embedded vision tokens into a pre-trained LLM. They are typically more suited for semantic understanding tasks.

For identifying visually-similar text, VLMs capable of encoding text directly are most desirable, mainly contrastive-based models. Leveraging text encoders avoids the usage of images for visual embeddings, reducing memory overhead, training complexity, and improving system-wide deployment. We conduct a baseline test to confirm the efficacy of contrastive-based models and select a pretrained VLM.

2.3 Methods

2.3.1 Model Architecture and Learning Objectives

Our proposed framework, VA-TE, builds upon the power of pretrained VLM embeddings to encode visual characteristics of text [44], rather than just semantic meaning. However, using a pretrained VLM without additional fine-tuning is insufficient, as the model faces problems when semantic-visual correlation fails. See Table 2.4 for reference.

Table 2.4: Examples of weak visual-semantic alignment.

Word 1	Word 2	Visual-Semantic Relationship Explanation
Doggy	Puppy	Visually different words with identical semantic meaning
Google	Goggle	Visually similar company names with different semantics

To refine the input embedding space, we train using four contrastive objectives: pairwise loss, triplet loss, InfoNCE, and Supervised Contrastive Loss (SupCon), each implemented based on their original formulations in the literature. The only notable deviation is our use of cosine similarity in the triplet loss instead of the traditional Euclidean distance—a change that yields equivalent results in practice due to prior embedding normalization.

Supporting these contrastive objectives, VA-TE follows a simple yet effective architecture (Fig. 2.1). Each input string is first passed through a frozen text encoder from a pretrained vision-language model, which produces a high-dimensional semantic embedding. These embeddings are then fed into a lightweight projection head, consisting of two shared linear layers with ReLU activation, designed to map semantic representations into a space that better reflects visual similarity. During training, the architecture operates as a Siamese network, receiving inputs and optimizing distances based on their label relationships, essentially pulling positive pairs closer together, while negative pairs are pushed apart. A prediction is made by computing the cosine similarity score between 2 input names, and classifying as a spoof if the score exceeds a predefined threshold.

2.3.2 Curriculum Learning

To strengthen the discriminative capability of the model, we employ three curriculum learning strategies. We use datasets of three difficulties: easy, medium, and hard, based on the separability of positives and negatives in the input (semantic) embedding space.

- **Manual Approach:** a fixed-schedule curriculum where the training process is divided into three equal stages by epoch, each stage increasing in example difficulty. This structured progression allows the model to first reinforce easily separable relationships in the input embedding space, before being challenged with hard negatives that closely resemble positives—forcing the model to learn fine-grained distinctions between visually-similar text.

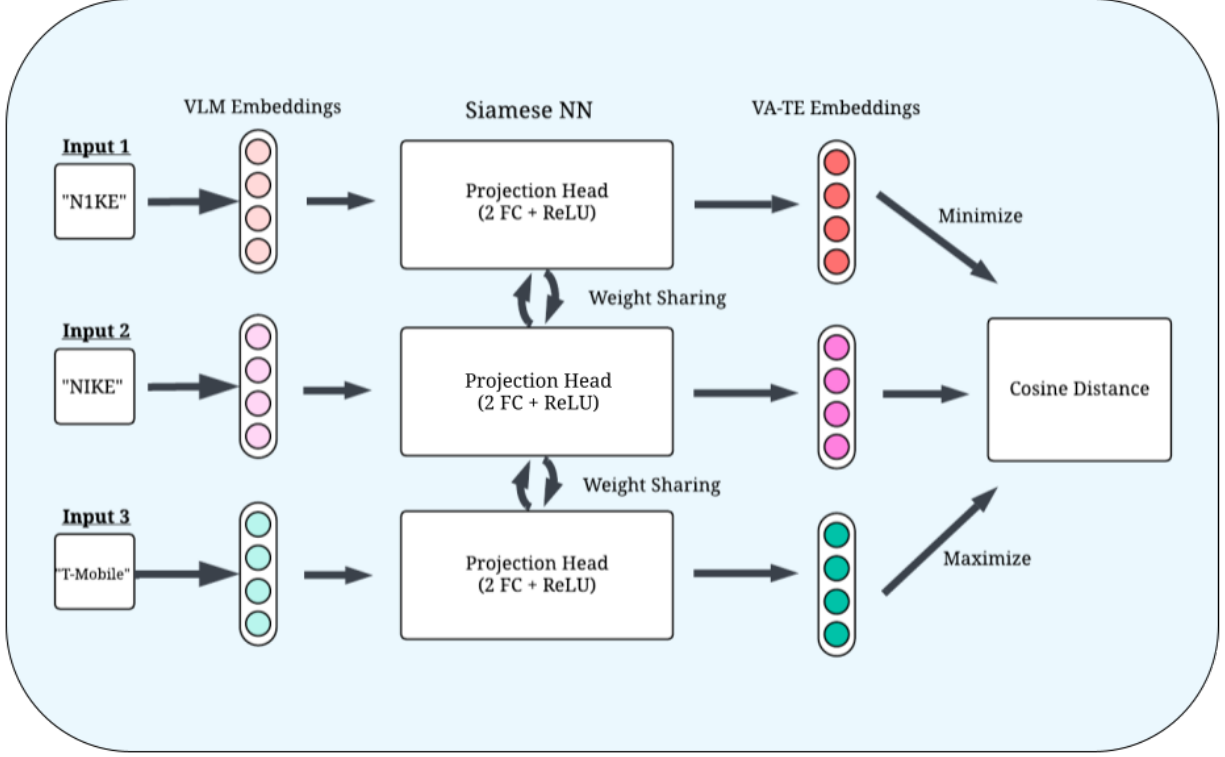


Figure 2.1: Overview of the VA-TE training architecture with light linear projector head.

- **Automated Approach:** instead of set stages, each epoch uses a training set with a mix of easy, medium, and hard samples. The ratio of samples from each difficulty level is determined by a scheduling function which dynamically adjusts these ratios as training progresses. The curriculum starts with a high proportion of easy samples and gradually shifts toward more hard negatives. This method follows a predefined curriculum, but provides smoother transitions than the manual approach, enabling more stable learning and potentially more effective adaptations to challenging examples. This scheduling strategy is defined by (2.5) and (2.6) and illustrated in Fig. 2.2.

$$t = \frac{\text{current epoch}}{\text{total \# of epochs}}, \quad c(t) = \cos(\pi t) \quad (2.5)$$

$$\begin{bmatrix} \text{easy}(t) \\ \text{medium}(t) \\ \text{hard}(t) \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(1 + c(t))(1 - t) \\ \frac{1}{2}[1 - (1 - 2t)c(t)] \\ \frac{1}{2}(1 - c(t))t \end{bmatrix} \quad (2.6)$$

- **Adaptive Bandit-Based Approach:** frames curriculum learning as a multi-armed bandit problem, where each difficulty level represents an arm of the bandit. At each epoch, the trainer balances exploration and exploitation by following an epsilon-greedy policy, selecting which dataset to use either at random or by estimated reduction in loss. This adaptive strategy allows the training process to dynamically focus on the most beneficial dataset, rather than adhering to a predefined schedule.

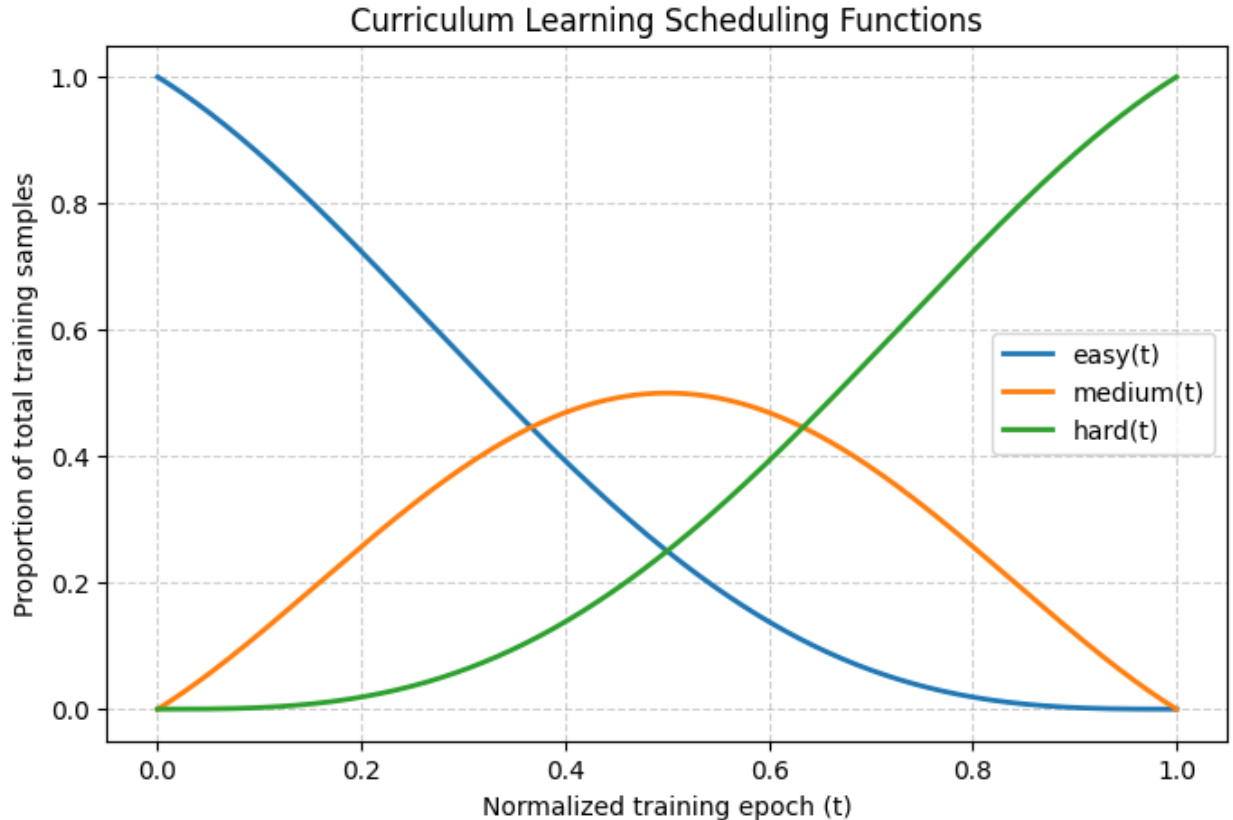


Figure 2.2: Automated curriculum scheduling function. Easy samples dominate early epochs, medium samples peak midway, and hard samples increase toward training end

2.3.3 Ensemble with Textual Features

Hard negatives pose a major challenge in spoof detection because they appear close to genuine company names in the VLM’s input (semantic) embedding space. Although contrastive objectives aim to separate these examples, hard negatives often remain near the anchor, especially when they share structural traits the model deems semantically relevant. Consequently, non-spoof names are frequently misclassified, exposing a key limitation of utilizing embedding-based methods alone: their reliance on transforming semantic similarity into visual-alignment can limit homoglyph detection accuracy.

Table 2.4 presents two representative examples where VLMs struggle due to conflicting visual and semantic cues. The first example, *Google* vs. *Goggle*, illustrates a **hard positive**. Although the two names are visually similar, they lie far apart in semantic space. As a result, even with fine-tuning, the VLM may fail to bring them sufficiently close in the embedding space to correctly classify the pair as a spoof, resulting in a false negative. In contrast, the second example, *Doggy* vs. *Puppy*, represents a **hard negative**. Here, the names are semantically similar but visually distinct. Because they are already close in semantic space, the model may fail to separate them adequately, leading to a false positive—incorrectly identifying a non-spoof as a spoof. These cases underscore the difficulty of spoof detection when visual and semantic signals diverge, motivating the addition of non-embedding string

methods that integrate strictly textual cues. This divergence becomes especially apparent in Table 2.5.

Table 2.5: Metric profiles of spoof-related examples. Spoofs and hard negatives are visually similar, but differ in string-based metrics.

Example Type	Cosine Similarity	Token Set Ratio	Levenshtein Distance
Spoof	High	High	Low
Hard Negative	High	Low	High
Hard Positive	Low	High	Low

Table 2.6: Average Levenshtein distance and token set ratio across spoof difficulty levels. Spoofs show lower distances and higher ratios.

Dataset	Spoof	Levenshtein Distance	Token Set Ratio
Easy	No	9.57	21.15
Easy	Yes	1.77	82.06
Medium	No	3.54	68.12
Medium	Yes	2.13	77.36
Hard	No	6.18	48.57
Hard	Yes	1.78	82.07

Table 2.6 shows that spoof and non-spoof pairs differ consistently across all difficulty levels: spoofs have lower Levenshtein distances and higher token set ratios. This distinction is especially pronounced in the hard subset, where embedding models struggle—spoofs average 1.78 in Levenshtein distance and 82.07 in token set ratio, while non-spoofs average 6.18 and 48.57, respectively. These results suggest that string-based metrics remain effective at separating spoof types that evade visual-semantic detection. Thus, to address the limitations of embedding models on visually deceptive cases, we adopt an ensemble approach that integrates string similarity with embedding-based representations, leveraging their complementary strengths to improve robustness.

Given a pair of strings (x_1, x_2) , we compute: (i) $s_{\text{VA-TE}} = \cos(\hat{z}(x_1), \hat{z}(x_2))$ where $\hat{z}(\cdot)$ are L2-normalized VA-TE embeddings; (ii) $r_{\text{TSR}} \in [0, 1]$, the Token Set Ratio (normalized to $[0, 1]$) capturing token overlap similarity; and (iii) $d_{\text{Lev}} \in \mathbb{N}$, the Levenshtein edit distance.²

We train a Gradient Boosting Classifier $g(\cdot)$ on the feature vector $\phi(x_1, x_2) = [s_{\text{VA-TE}}, r_{\text{TSR}}, d_{\text{Lev}}]$ to predict $p(\text{spoof} \mid x_1, x_2) = g(\phi(x_1, x_2))$. Unless noted otherwise, we use $n_estimators=100$, $max_depth=6$, $learning_rate=0.1$, $random_state=42$. Features are left unscaled (tree-based

²We use python-Levenshtein and RapidFuzz’s TokenSetRatio; distances are computed on raw strings.

models are scale-invariant). We fit g on the training split only, select the operating threshold τ on the validation split via Youden’s J , and report metrics on the held-out test set, ensuring data leakage does not occur.

Through this approach, $s_{\text{VA-TE}}$ encodes perceptual similarity learned from contrastive fine-tuning, while r_{TSR} and d_{Lev} are complementary, length-aware and edit-aware textual cues. A learned fusion lets the model weight each cue differently across regimes (string length, token overlap, character-level attacks), yielding a more robust homoglyph detector.

2.3.4 Parameter Tuning and Implementation Details

To effectively train our VA-TE model, we fine-tune several hyperparameters that influence overall model performance and convergence stability. These include learning rate, batch size, and internal layer size (for linear projection layers), as well as contrastive loss-specific temperature or margin parameters:

- **Margin**, used specifically for pairwise and triplet losses, defines the minimum required separation between negative and positive samples. This way, the model learns separation between difficult data points.
- **Temperature**, relevant only for InfoNCE and SupCon loss, scales similarity scores before softmax normalization. Lower temperatures produce sharper distributions with high contrast between positives and negatives, while higher temperatures produce less separation between contrasting data.

To determine optimal hyperparameters, we use **Optuna** ³, an optimization library that explores different regions of the hyperparameter space through probabilistic modeling.

Aside from accurate and efficient training, our implementation was designed specifically for scalability and deployment within practical systems. Other systems [14,15] require training from scratch on artificially-created images of text, running into dataset creation issues with inconsistent text sizing, along with scalability problems. We bypass these issues by building on pretrained VLM text encoders that don’t require image inputs. These advantages make our model more deployable in applications that involve text-based visual-similarity analysis. Through using a lightweight MLP projector and curriculum-based contrastive learning, our implementation lowers costs while simultaneously improving performance beyond other existing methods.

All code, datasets, and training configurations are publicly available in our GitHub repository at this link ⁴.

2.4 Experiments

2.4.1 Evaluation Task

To evaluate the effectiveness of VA-TE, we frame homoglyph detection of visually deceptive website domains [14] as the core downstream task. Within this setting, attackers register

³Documentation found at <https://optuna.org/>

⁴GitHub Repository available at <https://github.com/sashajh2/VA-TE>

domains that closely resemble those of legitimate sites, swapping, deleting, or inserting characters to create visually similar and deceptive alternatives. At its core, this evaluation allows us to determine whether the model’s learned embeddings are effective for textual visual similarity analysis.

2.4.2 Dataset

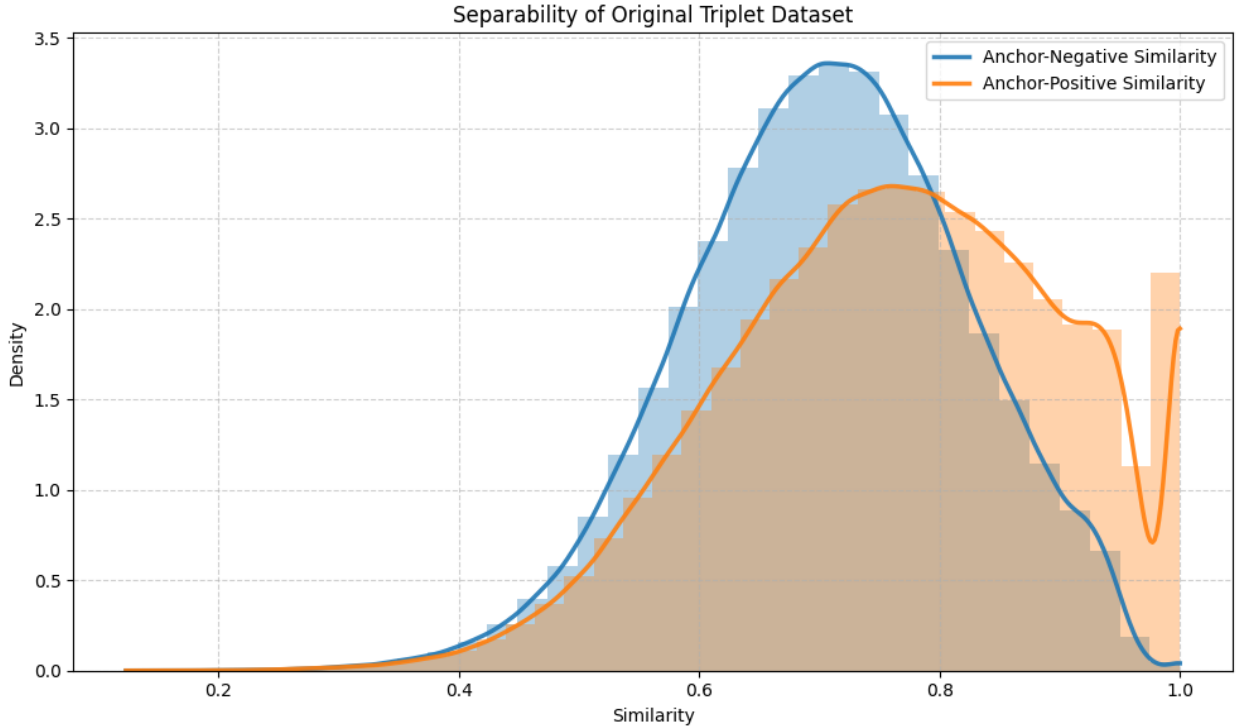


Figure 2.3: Separability of the triplet dataset without any modifications.

We use the publicly available dataset introduced by Woodbridge et al [14], constructed to support visual similarity learning for homoglyph detection in both domain and process names. We focus on the domain name dataset, training and evaluating using this dataset to benchmark our ensemble approach against prior works.

This original dataset was designed for contrastive learning using a Siamese CNN, where string pairs are rendered as grayscale images and labeled according to whether they represent a spoofing attack. The authors construct spoofed domains using predetermined homoglyph substitutions—targeted character swaps involving visually similar ASCII and Unicode characters. These manipulations yield domain pairs that are lexically distinct yet visually deceptive, reflecting the adversarial nature of homoglyph attacks.

The dataset is divided into three disjoint subsets—training (976,122 examples), validation (51,380 examples), and testing (256,886 examples)—for a total of 1,284,388 labeled pairs. Each example is a tuple containing a real domain name, a candidate spoof or unrelated domain, and a binary spoof label. We adapt this dataset to later generate triplets, along with InfoNCE- and SupCon-compatible examples.

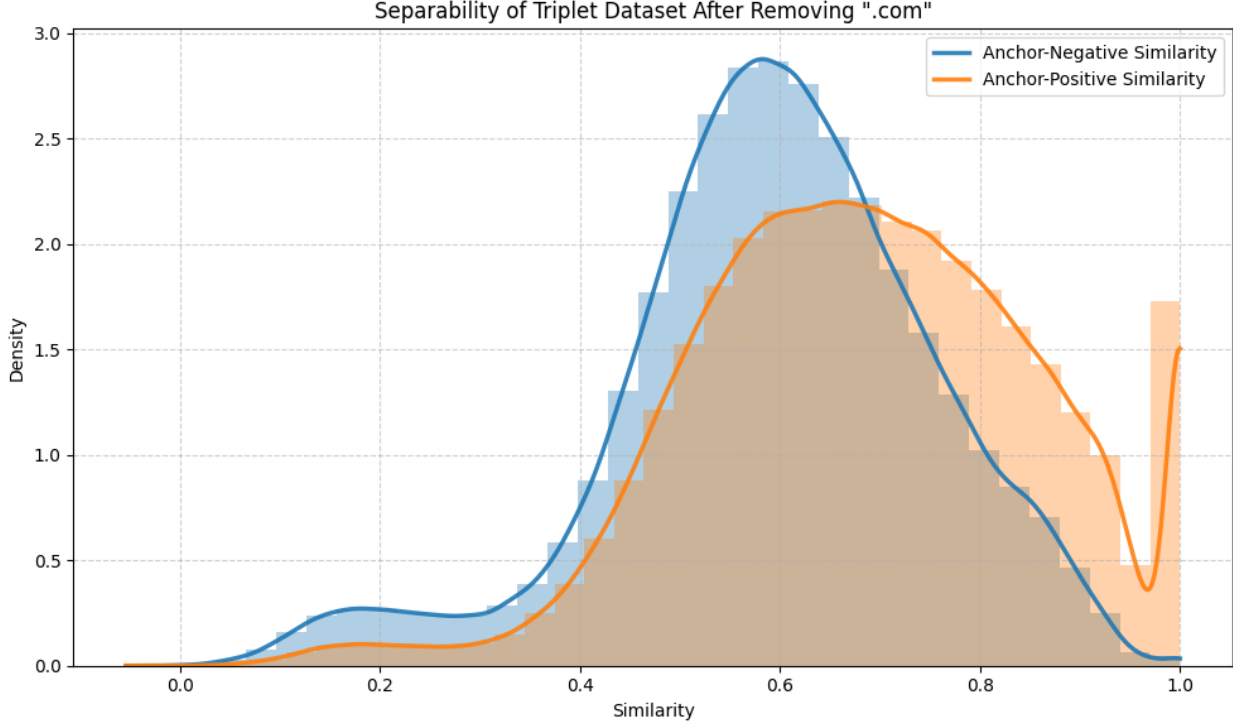


Figure 2.4: Separability of the triplet dataset after “.com” modification.

At inference time, the trained model takes in two domain names, generates their embeddings using the projector head, and computes their cosine similarity, which is then used to decide whether the input is a homograph attack.

Dataset Manipulations

Woodbridge et al. [14] express domain names in the format `example-domain.com`, introducing a constant “.com” suffix. This fixed token acts as a background signal in the input, possibly reducing the discriminative capabilities of the VLM’s pooled embedding and degrading performance. Based on insights from our separability analysis, we remove “.com” from all domain names to mitigate this effect (Table 2.7).

Table 2.7: Example domain transformation after removing .com.

Original	.com Removed
<code>example-domain.com</code>	<code>example-domain</code>

Our analysis indicates greater separability between the positive and negative pairs after removing .com.

Visually, Figure 2.3 and Figure 2.4 reveal a shift in the cosine similarity distribution of anchor-negative and anchor-positive examples after removing “.com,” with negative embeddings spreading further from the positive set. This shift is quantitatively supported by all

Table 2.8: Removing .com on separability of anchor–positive and anchor–negative cosine similarities.

Dataset	KL Divergence	JS Distance	Wasserstein
Original Dataset	0.1205	0.0347	0.0568
.com Removed	0.1442	0.0394	0.0893

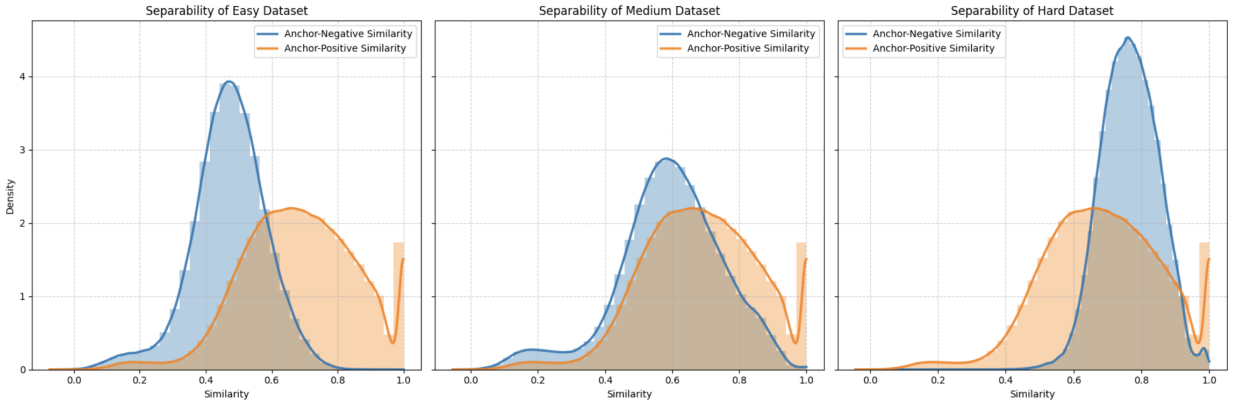


Figure 2.5: Similarity distribution between anchor–negative and anchor–positive pairs across easy, medium, and hard datasets.

three distributional metrics in Table 2.8. Wasserstein distance increased from 0.0568 to 0.0893 (+57%), indicating greater separation between anchor–positive and anchor–negative similarity distributions. KL divergence rose from 0.1205 to 0.1442 (+19.7%), while Jensen–Shannon distance grew from 0.0347 to 0.0394 (+13.5%). Together, these increases suggest improved contrast between positives and negatives post-removal, although the drop in overall similarity values may necessitate adjustment of decision thresholds.

For our curriculum learning experiments, we create datasets of varying difficulty by altering the negative sampling strategy. We define difficulty based on the prevalence of hard negatives, samples that are highly similar to the anchor in the input (semantic) embedding space. To quantify this, we compare the cosine similarity distributions of anchor–negative and anchor–positive pairs (see Fig. 2.5), using the latter as a reference to assess the prevalence of hard negatives.

To construct an easy dataset, we sample random negative examples, which tend to be more distinct from their respective anchors and thus yield a higher baseline separability than the original (medium difficulty) training dataset. For a hard dataset, we mine hard negatives, resulting in a much higher average anchor–negative cosine similarity. Training on these hard negatives forces the model to learn more subtle perceptual distinctions, improving its robustness against highly deceptive homograph attacks.

After removing “.com” from all domain names, mean negative similarity increases with dataset difficulty—0.4701 for the easy set, 0.5932 for the medium set, and 0.7658 for the hard set (Table 2.9). This trend reflects the growing challenge of distinguishing anchor–negative pairs as difficulty rises, with negatives in the hard set exhibiting similarity scores closer to those of positives—even surpassing them in some cases.

Table 2.9: Mean similarity values for anchor-negative and anchor-positive pairs after removing .com, across dataset difficulty levels.

Difficulty	Mean Negative	Mean Positive
Easy	0.4701	0.6825
Medium	0.5932	0.6825
Hard	0.7658	0.6825

2.4.3 Baseline Experiments

Vision-Language Models

Before applying training methods, we evaluate several pre-trained VLMs to determine which could serve most effectively as the backbone for VA-TE. Though these models were initially trained to capture semantic similarity across image-text pairs, we hypothesize that their internal representation may already encode relevant perceptual information, particularly regarding the visual appearance of text.

We test four prominent VLMs: CLIP, SigLIP, CoCa, and FLAVA. These models were selected to represent a range of architectures and initial training datasets. We hypothesize that dual-encoder designs, because they encode text and images separately, would create VA-TE best. For this reason, two different dual-encoder VLMs were tested: CLIP and SigLIP.

- **CLIP (Contrastive Language-Image Pretraining)** [7] uses contrastive loss to train on 400 million image-text pairs, minimizing the distance between matched pairs, and vice versa. Trained with softmax-normalized dot-product similarity and cross-entropy loss, CLIP excels at high-level semantic consistency. It uses a standard Transformer for text encoding purposes.
- **SigLIP (Sigmoid Loss of Language-Image Pretraining)**[47] is an adaptation of the CLIP framework that replaces softmax-based contrastive loss with a sigmoid-based pairwise loss, training the model using a binary cross-entropy over all pairwise matches within a batch. In contrast to CLIP’s softmax-based objective, which emphasizes relative similarity across a batch, SigLIP evaluates the relatedness of each image-text pair independently using a sigmoid-based loss. This pairwise formulation allows the model to learn finer-grained distinctions, which may be particularly helpful for preserving perceptual detail relevant to visual-similarity analysis.

Contrastive-based models are intuitively best-positioned for visual-similarity analysis. We verify this by testing select models from other architectures, mainly CoCa and FLAVA.

- **CoCa (Contrastive Captioners)**[48] is a hybrid encoder-decoder VLM that blend contrastive pretraining with image captioning objectives, introducing a captioning loss where the model learns to generate the associated text from the visual input.
- **FLAVA (Foundational Language and Vision Alignment)**[49] , a fusion-encoder model, is designed to jointly learn representations from both visual and textual modalities using a transformer backbone. Performing early fusion, concatenating visual

and language tokens then processing them together, FLAVA excels at understanding modality interactions.

Each model was evaluated in its frozen, zero-shot form, using its text encoder to generate embeddings for domain names, before being compared to each other using cosine similarity. With this baseline test, we evaluate how well each model’s native embeddings map visually-aligned text, without any additional fine-tuning. Selection criteria include ROC AUC, as well as accuracy, recall, and precision at Youden’s J .

SigLIP yielded the best baseline results as shown in Table 2.10, indicating its effectiveness for creating VA-TE. Thus, the remaining experiments conducted all train using SigLIP embeddings as input.

Table 2.10: *Baseline VLM test results.*

Model Name	ROC AUC	Accuracy	Precision	Recall
CLIP	0.6223	0.5447	0.7517	0.4357
CoCa	0.6798	0.5947	0.8721	0.4330
FLAVA	0.6092	0.5276	0.8806	0.3067
yellow!30 SigLIP	0.6874	0.6083	0.7735	0.5525

2.4.4 Fine-Tuning Experiments: Contrastive Loss and Curriculum Learning

We build upon the SigLIP text encoder by training two lightweight linear projection layers to map inputs into our target embedding space. To evaluate the effectiveness of different learning strategies, we test all pairwise combinations of four contrastive loss functions with each curriculum learning approach. As a control, we also fine-tune each loss function without curriculum learning using only original training data from [14], isolating the contribution of curriculum design.

2.4.5 Ensemble

After identifying the best VA-TE configuration (contrastive objective + curriculum strategy) using the validation set, we freeze the model and compute similarity scores s_{VA-TE} on the training, validation, and test splits. We then train a Gradient Boosting Classifier using three input features: s_{VA-TE} , Token Set Ratio, and Levenshtein distance. The decision threshold is tuned on the validation set, and test performance is evaluated via ROC-AUC. To assess the benefit of ensembling, we also report ablation results for using VA-TE alone and string-based features alone.

2.4.6 Evaluation Metrics

Our primary evaluation metric is ROC-AUC, which enables direct comparison with previous works [14,15] and provides a robust summary of the model’s ability to distinguish visually similar text. We additionally report Accuracy, Precision, and Recall to offer a more detailed view of performance under practical performance metrics.

2.5 Results

2.5.1 Baseline Performance

To assess the effectiveness of traditional string matching approaches, we evaluated Levenshtein Distance and Token Set Ratio on the spoof detection task. As shown in Table 2.11, Token Set Ratio slightly outperforms Levenshtein Distance in ROC-AUC (0.8350 vs. 0.8137), indicating that it captures a slightly larger range of visually-similar text. However, these results aren’t viable compared to existing methods, despite scalability advantages. This motivates the usage of VLMs for better learned representations of visual similarity.

Table 2.11: *Baseline String Matching Results*⁵

Metric	ROC AUC	Accuracy	Precision	Recall
Levenshtein Distance	0.8137	0.8170	0.9046	0.5450
Token Set Ratio	0.8350	0.7616	0.9110	0.6973

2.5.2 VA-TE Performance

Across 20 different VA-TE training configurations, we explored the interaction between contrastive loss functions and curriculum learning strategies. As shown in Table 2.12, the pairwise contrastive loss consistently achieved the highest ROC-AUC scores, reaching 0.95 across nearly all curriculum variations. For triplet loss, scores hovered around 0.91–0.92 regardless of curriculum. InfoNCE and SupCon performed competitively (0.92–0.94), with Automated Curriculum slightly boosting SupCon to 0.94. These results indicate that pairwise loss is robust to sampling difficulty, triplet remains weaker overall, while InfoNCE and SupCon show modest gains from curriculum learning.

Taken together, these findings underscore VA-TE’s competitive performance and practical deployability compared to SOTA alternatives. Nevertheless, vision-language models alone are limited in capturing the textual perturbations common in homoglyph attacks. This motivates an ensemble approach, which corroborates VA-TE similarity scores with Levenshtein distance and fuzzy string matching.

⁵Accuracy, precision, and recall are computed at the Youden’s J threshold.

Table 2.12: *Validation and test ROC-AUC across VA-TE training strategies.*

Strategy	Pair	Triplet	InfoNCE	SupCon
Medium Negatives	0.95	0.92	0.92	0.92
Hard Negatives	0.94	0.91	0.93	0.93
Manual Curriculum	0.95	0.92	0.93	0.92
Automated Curriculum	0.95	0.93	0.93	0.94
Bandit Curriculum	0.95	0.92	0.92	0.92

2.5.3 Ensembles

To evaluate the benefits of combining vision-language and string-based features, we constructed an ensemble model using the best VA-TE configuration alongside traditional string similarity metrics. As detailed in Table 2.13, the ensemble achieved a ROC-AUC of 0.98, outperforming both the VA-TE model alone (0.95) and the string-based metrics alone (0.89). This significant improvement highlights the complementary strengths of visual-semantic embeddings and token-level string comparisons, especially in detecting visually deceptive text where either modality alone may be insufficient.

Table 2.13: *ROC-AUC for Ensemble Strategies.* “String Metrics” refers to the combination of Edit Distance and Token Set Ratio.

Ensemble Strategy	ROC AUC
String Metrics	0.89
VA-TE + String Metrics	0.98

2.5.4 Comparison to State-of-the-Art

Compared to prior SOTA methods, our VA-TE framework alone achieves highly competitive results while maintaining a lightweight and scalable architecture. As shown in Table 2.14, the VA-TE ensemble attains SOTA results with a ROC-AUC of 0.98, surpassing traditional models that rely heavily on domain-specific heuristics or large-scale neural architectures. Furthermore, while other deep learning approaches require substantial training time and preprocessing, VA-TE benefits from compact linear projection heads over pretrained encoders, enabling easier deployment beneficial for real-world applications that demand both accuracy and efficiency. These results demonstrate that a semantically grounded representation can be effectively translated into a visually aligned embedding space, enabling high performance on downstream tasks.

Table 2.14: *Comparison of proposed methods to state-of-the-art models on the ROC-AUC metric.*

Method	ROC AUC
VA-TE	0.95
VA-TE + String Metrics	0.98
Siamese-GRU [15]	0.98
Siamese-LSTM [15]	0.97
Siamese-CNN [14]	0.97
Siamese-CNN [15]	0.93

2.6 Discussion

2.6.1 Performance Analysis

Our experiments highlight important insights into the efficacy of VA-TE and the role of different training configurations.

Across all settings, pairwise contrastive loss consistently achieved the strongest performance, reaching a ROC-AUC of 0.95. This result is consistent with the SigLIP encoder’s original pairwise training objective [47] and avoids the margin sensitivity issues that hinder alternative loss formulations. In particular, triplet loss plateaued around 0.91-0.92, underperforming pairwise. This is likely due to difficulties in margin tuning and limited batch sizes restricting effective negative sampling. InfoNCE and SupCon achieved competitive results, but showed greater dependence on curriculum learning to achieve optimal performance.

The impact of curriculum learning varied according to the loss function used. For pairwise and triplet losses, curriculum learning yielded negligible improvements, suggesting that these objectives are relatively robust to changes in negative sampling difficulty. In contrast, InfoNCE and SupCon gained modest but consistent benefits from Automated Curriculum, while manual and bandit-based approaches showed little added value. Across the four loss functions, the Automated Curriculum was consistently the strongest performer, matching the best result or achieving the top score outright, providing smoother transitions and more stable convergence than alternative curricula.

When compared against prior benchmarks, VA-TE achieved its best standalone performance with pairwise loss and automated curriculum (ROC-AUC = 0.95). More importantly, when fused with string-based similarity metrics such as Levenshtein distance and Token Set Ratio, the ensemble attained state-of-the-art performance with a ROC-AUC of 0.98. This surpasses or matches the strongest image-based approaches, including Siamese-CNN and Siamese-GRU models [14,15]. Unlike those methods, VA-TE operates directly on text rather than rendered images, providing substantial advantages in scalability, memory efficiency, and deployment simplicity while maintaining competitive accuracy. These findings underscore both the robustness of pairwise contrastive learning in visually-aligned text embeddings and

the complementary value of combining embedding-based and string-based features.

2.6.2 Advantages of VA-TE

Scalability Benefits over Image-Based Approaches

VA-TE offers substantial advantages over existing image-based homograph detection methods in terms of computational efficiency and memory requirements. Previous state-of-the-art approaches [14,15] require rendering each string as a binary image before training and inference. Specifically, Woodbridge et al. [14] render strings as 150×12 pixel grayscale images. The associated datasets contain 1,284,388 labeled pairs, requiring the generation and storage of over 2.5 million images. In contrast, VA-TE operates directly on text strings, eliminating image rendering entirely and reducing memory overhead by orders of magnitude.

Beyond storage considerations, image-based methods introduce additional computational overhead during both training and deployment. Each inference requires: 1) String-to-image rendering using specific fonts and sizing parameters, 2) Image preprocessing and normalization, and 3) Forward pass through a convolutional neural network designed for image input. VA-TE eliminates steps 1 and 2 entirely, requiring only a lightweight forward pass through the frozen VLM text encoder followed by the projection head. This streamlined pipeline enables real-time deployment in production environments where latency and computational resources are critical constraints.

Deployment Simplicity and Robustness

VA-TE’s text-only design significantly simplifies deployment by avoiding the rendering used by image-based methods, which depend on parameters like font choice, size, anti-aliasing, and background color. Inconsistencies in these parameters can degrade model performance and require pipeline maintenance. By relying on pretrained VLM text encoders, VA-TE avoids these issues entirely, enabling consistent performance across environments without graphics libraries or image processing, reducing the risk of deployment failures.

Additionally, image-based approaches impose artificial constraints on input string length due to fixed image dimensions. CNN-based models often require truncating or padding strings to fit a predetermined image size (e.g., the 25-character limit in [14]), introducing a source of information loss or distortion. By contrast, VA-TE can process strings of arbitrary length limited only by the underlying VLM’s context window, typically accommodating hundreds of characters.

Furthermore, VA-TE leverages the robustness of large-scale pretrained models. SigLIP and similar VLMs are trained on massive, diverse datasets and demonstrate strong generalization capabilities. By building on these established representations rather than training from scratch on synthetically generated images, VA-TE inherits the robustness and stability of the underlying foundation model.

2.6.3 Future Work

Application to Different Text Domains

Extending beyond English, visually deceptive text persists in other, more complex writing systems. Detecting homoglyphs in languages such as Chinese, Arabic, or Cyrillic would require more fine-grained character distinctions and introduces additional challenges, including font-based homographs and diacritic variations.

Alternatively, VA-TE can serve as a verification layer for Optical Character Recognition (OCR) pipelines, particularly in document digitization. Recent literature has shown that OCR systems are susceptible to misclassifying visually similar characters—particularly in difficult-to-distinguish languages like Arabic [50]. Embedding-based homoglyph detection could identify and automatically correct these OCR-induced misclassifications, providing vast applications in security-critical contexts, including bank check processing, passport scanning, and historical document transcription.

Enhancing Text Embeddings via Perceptual Decoding

While our current approach leverages the joint text-image embedding space of pretrained VLMs, it relies exclusively on the *text encoder* to generate representations of input strings. These embeddings, though semantically rich and convenient for scalable deployment, are aligned with the final-layer outputs of the VLM’s text encoder and therefore may not capture perceptual features necessary for fine-grained homoglyph detection.

To address this limitation, we propose an architectural extension that grounds text embeddings in low- and mid-level visual features extracted from rendered images of the input text. Specifically, we introduce a lightweight decoder $f_d : \mathbb{R}^{d_{text}} \rightarrow \mathbb{R}^{d_{vis}}$ trained to map text embeddings to multi-scale image features, extracted from shallow, intermediate, and deep layers of the VLM’s vision encoder. During training, this decoder learns to reconstruct visual features from text embeddings alone, encouraging the text representations to encode subtle typographic details. Crucially, once trained, the decoder becomes part of the permanent inference pipeline, transforming text embeddings into visually-grounded representations without requiring image inputs at deployment time. We can approach this training in two ways:

1. **Preprocessing module:** Pre-train the encoder to enhance the text embedding space prior to standard contrastive training.
2. **Joint Optimization:** Simultaneously train the decoder and contrastive objectives, allowing the text encoder to adapt its representations for both visual reconstruction and similarity learning.

This formulation offers a promising path for improving visual discrimination in spoof detection while preserving the efficiency and deployment simplicity of text-only inference.

Enhanced Model Architectures and Training Strategies

Future architectures could benefit from delayed weight update strategies, where gradients are accumulated over multiple batches before parameter updates. Given the relatively lightweight

projection head (two linear layers) and the noisy nature of contrastive learning objectives, gradient accumulation could provide more stable learning signals and improved global optimization trajectories. This approach would be particularly valuable when computational constraints limit batch sizes, allowing the model to approximate the benefits of larger-batch training without additional memory overhead.

Focusing on structural improvements, prior literature has demonstrated the effectiveness of recurrent architectures such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) for homoglyph detection [15]. Future work could explore integrating these recurrent models with VLM-based approaches to better capture sequential dependencies and positional patterns in character embeddings.

Chapter 3

Learning What’s Real: Intrinsic Authenticity Detection for Generalization Across Deepfake Methods

3.1 Introduction

Deepfake generation technology has been advancing at an unprecedented pace, posing significant threats to systems dependent on identity verification and public trust. Early deepfakes relied on simple face-swapping techniques using Generative Adversarial Networks (GANs), but modern diffusion models are now able to produce synthetic media with photorealistic quality that is increasingly difficult to distinguish from authentic content. This rapid evolution in generation capabilities and methods has exposed critical vulnerabilities in existing detection approaches. Current deepfake detection methods face two fundamental limitations. First, many approaches train exclusively on synthetic data generated by a single model, learning to identify artifacts specific to that generation method [cite]. While this strategy achieves reasonable performance on data from within the training distribution, it drastically underperforms when confronted with novel generation techniques. Given the accelerating pace of advancement in this field, this poses a critical weakness for identifying authentic media. Second, alternative approaches attempt to leverage embeddings from various modalities, namely audio and video features, to distinguish real from synthetic media (cite). However, these learned representations predominantly encode content-related information rather than the subtle artifacts and inconsistencies that reliably indicate synthetic generation. These limitations motivate our approach: disentangling authenticity-related features from those encoding identity and content within pretrained representations, isolating the signals most relevant for deepfake detection. By explicitly separating these feature types, we aim to reduce the influence of extraneous information, which may be similar across both real and fake media, while amplifying detection-relevant signals that generalize across generation methods. Our central hypothesis is that this disentanglement will enable learned representations to cluster authentic and synthetic media into distinct regions of a latent space, achieving robust generalization to previously unseen generation techniques. This approach addresses both the overfitting problem of technique-specific detectors and the content-encoding limitation

of existing representation-based methods, offering a more principled path toward reliable, generalizable deepfake detection.

3.2 Related Works

3.2.1 Deepfake Generation

Generative Adversarial Networks

Along with other developments in AI, synthetic media generation has rapidly evolved, with each subsequent generative model family producing increasingly realistic samples. Early deepfake methods relied on Generative Adversarial Networks (GANs), which introduced an adversarial generator-discriminator framework that quickly became the leading paradigm for synthetic media creation [51]. GAN models were soon applied across modalities, including audio, where early works like WaveNet demonstrated that adversarially trained architectures could generate highly realistic audio samples [52]. Significant progress in GANs was driven by Karras and colleagues through a sequence of architectural innovations. ProGAN [53] iteratively increases the size of both the generator and discriminator, which stabilizes training and enables high-resolution face syntheses with improved diversity. StyleGAN [54] reorganized the generator around a style-based architecture that first maps the input latent embedding into an improved latent space and then injects this representation into each layer of the generator. This allows the generator model to naturally disentangle coarse structure, mid-level semantic features, and fine-grained texture details. Parallel to these advances in generative modeling, face-swap models such as Face2Face [55] and SimSwap [56] focused on identity transfer rather than full-frame synthesis, enabling real-time and high-quality manipulations by conditioning on target facial expressions and landmarks. Despite these advances, GAN-generated media often contains recognizable artifacts that remain detectable by modern detection systems [57]. These limitations, combined with the training instability of GANs, helped motivate the shift toward diffusion-based models that offer greater stability and consistently produce more realistic synthetic media.

Diffusion Models

Diffusion models, first introduced in 2015 [58] based on ideas from statistical physics, function by gradually adding noise to data and then learning a reverse denoising process to recover the underlying signal. This procedure allows diffusion models to more accurately model the full data distribution and avoid the mode-collapse issues of GANs. By 2020-2021, they emerged as the state-of-the-art class of generative models with the invention of foundational architectures such as Denoising Diffusion Probabilistic Models (DDPMs), which showed dramatic improvements in image quality and diversity [59]. Compared to GANs, diffusion models produce fewer artifacts, and instead suffer from issues like temporal inconsistency in video outputs [57]. Current diffusion-based video generation models, including OpenAI’s Sora and Google’s Veo3, now produce photorealistic video and audio, motivating the development of detection techniques that remain robust to rapidly advancing generative models.

3.2.2 Deepfake Detection Approaches

Existing deepfake detection methods can be broadly categorized into forensic and machine learning based approaches, each with distinct strengths and limitations [57].

Forensic Methods

Forensic techniques use signal processing, hand-crafted features, and domain knowledge to detect manipulation artifacts. They typically analyze frequency domain statistics and resolution inconsistencies [60], along with surface warping [61], inconsistent lighting, and other errors resembling those found in related video-processing tasks such as face tracking and editing [62]. Biometric-based forensic methods exploit the inability of generators to accurately model biological signals, such as heart-rate patterns [63–65], lip-sync mismatches [66], and joint jaw-ear motion dynamics [67]. While effective when visible artifacts are present, forensic methods fail on modern generators that produce high-quality, high-fidelity outputs with minimal artifacts. They are also vulnerable to video compression and low-resolution inputs, which disrupt or erase the subtle forensic signals these methods rely on [57].

Machine Learning Methods

Machine learning-based detectors move beyond forensic cues by learning discriminative features directly from data. By operating in latent spaces that capture richer semantic and structural information, these models can leverage large datasets, reduce information loss, and capture subtle inconsistencies that may not be visible in pixel or frequency space features. Early CNN-based detectors such as XceptionNet (Rössler et al. 2019) achieved strong performance by capturing the spatial features of manipulation artifacts. Subsequent works introduced attention mechanisms to emphasize both localized and global cues, as well as temporal modeling via 3D CNNs, ConvLSTMs, and transformer-based encoders to detect temporal inconsistencies across audio and video streams. As deepfakes have improved in quality, unimodal visual analysis has proven insufficient, leading to multimodal methods that jointly model audio and video correlations. These approaches learn a joint embedding space where inconsistent lip motion, rhythm, and emotions can be detected. Despite these advances, ML-based detectors still struggle to generalize to unseen generative models [find citation?]. A key reason is representation entanglement: identity, semantic content, temporal patterns, and authenticity cues are all mixed together within the embedding space. This entanglement encourages detection models to rely on identity-specific or dataset-specific shortcuts, degrading performance under distribution shift and novel generation methods. This limitation motivates methods that explicitly disentangle authenticity-related information from identity and other confounding factors.

3.2.3 Disentangled Representation Learning

While disentanglement has been widely studied in generative modeling [cite], its application to deepfake detection remains limited. In discriminative settings, separating task-relevant signals from confounding factors improves robustness and reduces reliance on dataset-specific or identity-specific correlations. This is especially applicable to deepfake detection, where

standard models often entangle who appears in the video with whether the video is authentic and simultaneously overfit to the specific generative models present in their training set. As a result, detectors rely on identity shortcuts or generator-specific artifacts rather than manipulation-invariant cues, leading to significant failures under distribution shift. Recent anomaly detection approaches address this by decomposing latent feature spaces into homogeneous (consistent across all real images) and heterogeneous (image-specific) components [68]. These methods generalize better to unseen generative models because they learn intrinsic properties of real images rather than generator-dependent artifacts. However, they do not disentangle identity information from authenticity information, leaving a critical source of shortcut learning unaddressed. Our approach directly targets this gap through disentangled, real-video-driven pre-training. We leverage the assumption that all real videos share a common authenticity signal that is independent of content, identity, or scene variations. To isolate this signal, we conduct pre-training exclusively on real videos and learn two complementary latent subspaces:

1. **Authenticity-preserving subspace:** A projection head maps embeddings of real videos into a low-variance manifold capturing the statistical commonalities shared across all authentic footage. This manifold represents the intrinsic “realness” structure of authentic spatiotemporal data.
2. **Content/Identity subspace:** A second projection head, constrained to be orthogonal to the authenticity-preserving one, captures all remaining variation (identity, semantics, motion style, context). We train this subspace contrastively on augmentations of the same video to ensure it captures expressive, person-specific variation.

The orthogonality constraint minimizes mutual information between the two latent codes, encouraging the authenticity branch to encode only manipulation-relevant signals, and the identity branch to absorb variation unrelated to authenticity. At inference time, the authenticity-preserving projection head serves as an anomaly detector. Real videos project close to the learned low-variance manifold, while synthetic videos, which lack the intrinsic real-video commonalities, deviate sharply from it. This combines principles from disentangled representation learning, anomaly detection, and commonality learning to model authenticity directly, without needing examples of every manipulation type. Finally, to evaluate the learned spaces, we adopt classifier-free embedding metrics to assess the quality of disentanglement and robustness without relying solely on downstream classifier tuning. These metrics allow for fair comparison across representation spaces and directly reflect the geometric structure induced by our method.

3.3 Methods

3.3.1 Overview: Two-Stage Detection Framework

We frame deepfake detection as a temporal binary classification problem, where the goal is to estimate the authenticity of each segment of an audio-video sequence to enable precise temporal localization. Our approach decomposes this task into two stages: (1) disentangled

representation learning, which first isolates authenticity-related signals from content and identity features, and (2) temporal classification, which models sequential dependencies to produce segment-level predictions. This design leverages the fact that pretrained embeddings already contain meaningful authenticity cues, but also encode substantial variation due to identity and scene content. By explicitly separating these components, the authenticity projection learns a low-variance manifold shared across all real videos, enabling the model to isolate robust signals for deepfake detection. The orthogonal projection simultaneously captures identity and content structure by contrasting content-invariant augmentations, learning representations that remain useful for complementary downstream tasks such as biometric verification.

3.3.2 Disentangled Representation Learning

Dual Projection Heads

Our representation learning architecture consists of a shared encoder f_{enc} followed by two projection heads, f_{auth} and f_{id} . Given an input frame x , the encoder produces an embedding $z = f_{\text{enc}}(x)$, which is projected into two complementary subspaces:

$$z^{\text{auth}} = f_{\text{auth}}(z) \quad : \text{captures authenticity-related signals.} \quad (3.1)$$

$$z^{\text{id}} = f_{\text{id}}(z) \quad : \text{captures identity and content information.} \quad (3.2)$$

The encoder f_{enc} may be any pretrained visual or audio-visual model (Section 3.4.2). Both projection heads are implemented as two-layer MLPs with ReLU activations and output dimension $d = 128$.

This dual-head structure enables explicit supervision of each subspace using tailored objectives, while the shared encoder ensures computational efficiency and a unified representation from which both identity and authenticity can be decoded. Our design is motivated in part by recent work showing that separating complementary factors in the representation space can improve generalization in media forensics [68].

Orthogonality Constraint

To enforce independence between identity and authenticity representations, we introduce an orthogonality constraint that penalizes correlation between z^{id} and z^{auth} . Formally, for a batch of N samples, we minimize:

$$\mathcal{L}_{\text{orth}} = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N |\text{sim}(z_i^{\text{id}}, z_j^{\text{auth}})| \quad (3.3)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity. This constraint is inspired by information bottleneck principles and decorrelated representation learning methods such as Barlow Twins and VICReg, and encourages the two projection heads to extract complementary, non-redundant information from the shared encoder. **NEED CITATION HERE!**

Intuition: If identity and authenticity are orthogonal in the learned embedding space, then knowing someone’s identity should provide no information about whether their video

is real or fake. This property is critical for generalization: a model that disentangles these factors should not memorize identity-specific deepfake patterns (e.g., “person A is always fake in the training set”) but instead learn identity-agnostic authenticity cues.

Training Objectives

We train the dual projection heads with two complementary objectives: (1) a prototypical contrastive loss for identity learning, and (2) a variance minimization loss for authenticity learning.

Identity Learning via Prototypical Contrastive Loss The identity head f_{id} is trained to cluster embeddings based on content similarity, regardless of authenticity. We leverage the augmentation structure of our datasets: in AVDeepfake-1M++, multiple augmented versions of the same source video share identical content but differ in noise/perturbations, while in ShareVeo3, consecutive frames within a video share semantic content.

We adopt prototypical contrastive learning to make training computationally efficient ($O(n)$ rather than $O(n^2)$ pairwise comparisons in standard contrastive learning). For each “content group” (augmentations of the same source video or frames from the same video), we compute a prototype as the mean embedding:

$$c_k = \frac{1}{|G_k|} \sum_{i \in G_k} z_i^{\text{id}} \quad (3.4)$$

where G_k is the set of samples belonging to content group k . The prototypical loss encourages embeddings to be close to their assigned prototype and far from other prototypes:

$$\mathcal{L}_{\text{proto}} = -\frac{1}{N} \sum_{i=1}^N \log \left[\frac{\exp(-d(z_i^{\text{id}}, c_k)/\tau)}{\sum_j \exp(-d(z_i^{\text{id}}, c_j)/\tau)} \right] \quad (3.5)$$

where $d(a, b)$ is Euclidean distance, τ is a temperature hyperparameter, and the sum is over all prototypes in the batch.

Authenticity Learning via Variance Minimization The authenticity head f_{auth} is trained using an anomaly detection approach: we train only on real videos, forcing their embeddings to collapse into a tight cluster. The intuition is that real videos share common authenticity characteristics (e.g., natural sensor noise, consistent photometric properties), whereas fake videos exhibit diverse artifacts depending on the generation method. By minimizing variance among real embeddings, we create a compact “real manifold” in the z^{auth} space, leaving room for fake videos to scatter outside this region.

Formally, for a batch of real samples (authenticity label = 1), we minimize:

$$\mathcal{L}_{\text{var}} = \frac{1}{N_{\text{real}}} \sum_{i: y_i=1} \|z_i^{\text{auth}} - \mu_{\text{real}}\|^2 \quad (3.6)$$

where μ_{real} is the mean real embedding in the batch.

Joint Optimization The total loss for the representation learning stage is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{proto}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}} + \lambda_{\text{orth}} \mathcal{L}_{\text{orth}} \quad (3.7)$$

where λ_{var} and λ_{orth} are hyperparameters balancing the three objectives. We set $\lambda_{\text{var}} = 0.5$ and $\lambda_{\text{orth}} = 0.1$ based on validation performance (see Section 3.5 for ablations).

3.3.3 Temporal Classification

Architecture

After learning disentangled representations, we train a temporal classifier to model sequential dependencies and make frame-level authenticity predictions. The input to this stage is a sequence of authenticity embeddings $\{z_1^{\text{auth}}, z_2^{\text{auth}}, \dots, z_T^{\text{auth}}\}$ extracted from T consecutive frames of a video (we use $T = 32$ frames with 2 fps sampling).

We use a Transformer encoder with $L = 4$ layers, $d_{\text{model}} = 128$, and $h = 4$ attention heads. The Transformer processes the full sequence and outputs contextualized embeddings $\{h_1, \dots, h_T\}$. Each contextualized embedding h_t is passed through a frame-level classification head (a 2-layer MLP with sigmoid output) to predict the probability that frame t is fake:

$$p(\text{fake} \mid \text{frame } t) = \sigma(\text{MLP}(h_t)) \quad (3.8)$$

Design rationale: Per-frame predictions enable real-time detection and localization of injection attacks (where only part of a video is manipulated, as in AVDeepfake-1M++). The Transformer’s self-attention mechanism allows each frame to attend to context from neighboring frames, capturing temporal inconsistencies that may indicate manipulation (e.g., abrupt transitions at injection boundaries).

Training Strategy

We train the temporal classifier with frame-level binary cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{T} \sum_{t=1}^T [y_t \log p_t + (1 - y_t) \log(1 - p_t)] \quad (3.9)$$

where $y_t \in \{0, 1\}$ is the ground-truth label for frame t (0=real, 1=fake). For fully real videos (e.g., from AVDeepfake-1M++ real set), all frames have $y_t = 0$. For fully fake videos (e.g., ShareVeo3), all frames have $y_t = 1$. For partial fake videos (AVDeepfake-1M++ fake set), $y_t = 1$ only for frames within the manipulated temporal segment, and $y_t = 0$ otherwise.

Important design choice: The representation learning stage (Section 3.3.2) is frozen during temporal classifier training. This two-stage approach ensures that the disentanglement objectives are not corrupted by the classification loss, which might otherwise push z^{auth} to encode identity information if it helps discriminate fake videos in the training set.

3.4 Experiments

3.4.1 Datasets

We train and evaluate our method using two complementary datasets: AVDeepfake-1M++ and ShareVeo3, which together provide diverse coverage of deepfake generation methods and manipulation scenarios. AVDeepfake-1M++ focuses on face-swap and voice-cloning manipulations applied to real source footage, while ShareVeo3 and OpenAI Sora2 represent end-to-end text-to-video generation models that synthesize entire videos from scratch. These generative models pose a particularly severe threat to identity verification systems, as they can generate videos targeting specific faces and voices from minimal training data, often just a few seconds of footage or publicly available social media content. We use ShareVeo3 for training to expose our model to diffusion generation, and evaluate out-of-distribution generalization on Sora2, a state-of-the-art video generation model not represented in our training data [69].

For all datasets, we segment videos and audio into 0.15-second segments and generate audio and video embeddings for each segment using pretrained encoders. These embeddings are stored in a serverless PostgreSQL database for efficient retrieval during training and evaluation. The rationale for this temporal segmentation granularity is discussed in Section 3.4.2 (Embedding Selection), where we analyze the trade-off between temporal resolution and computational efficiency.

AVDeepfake-1M++

AVDeepfake-1M++ [70] is a large-scale audio-visual deepfake dataset containing over 2 million videos spanning 7,109 unique subjects across training, validation, and test splits. The dataset employs diverse generation methods for both visual and audio manipulation, including:

- **Visual deepfakes:** WAV2LIP (100% of visual manipulations in base dataset), TalkLip (100% in base), and an expanded set in AVDeepfake-1M++ including TalkLip (65.9%), Diff2Lip (22.9%), and LatentSync (11.2%)
- **Audio deepfakes:** SV2TTS (100% in base dataset), VITS (66.6% in base), YourTTS (33.4% in base), and an expanded set in AVDeepfake-1M++ including VITS (38.8%), YourTTS (38.0%), F5TTS (9.9%), and XTTSv2 (13.2%)

From this large-scale dataset, we construct our training subset by selecting 241 source videos from the training split, each with approximately 10 augmented versions created through synthetic perturbations. Our preprocessing pipeline extracts embeddings at 0.15 second intervals yielding 231,889 total embeddings. This subset has two key structural properties that we leverage for representation learning:

- **Multiple augmentations per source:** Each source video is augmented with diverse perturbations including Gaussian blur, salt-and-pepper noise, compression artifacts, color quantization, motion blur, and others. Crucially, all augmentations of the same

source share identical underlying content (same subject and audio-visual semantics), differing only in low-level signal properties induced by perturbations and authenticity signals. This provides strong supervision for identity learning: augmented versions of the same source should cluster tightly in z^{id} space regardless of their authenticity label or applied perturbations.

- **Partial fake injections:** Fake videos in AVDeepfake-1M++ are created by injecting synthesized audio or video segments into real source material, with temporal injection boundaries annotated in the video metadata. This enables frame-level evaluation and tests the model’s ability to localize temporal manipulation boundaries, a capability lacking in most video-level detectors that only provide binary video-level predictions.

After preprocessing and embedding extraction, our training subset contains: 225,843 fully real embeddings (covering original source videos and real augmentations with perturbations but no deepfake manipulation) 2,394 fully fake embeddings (videos with complete audio and/or visual synthesis throughout) 3,652 partial fake embeddings (videos with temporal injection, where only segments are manipulated) We address class imbalance through balanced batching during training: each batch contains equal numbers of real and fake samples, sampled uniformly across content groups (where a content group is defined as all augmentations derived from the same source video).

For prototypical learning, we define content groups by source video identity. All augmentations of the same source video, regardless of perturbation type or authenticity, belong to the same content group and should share similar z^{id} representations. We use all available augmentations rather than subsampling to maximize the diversity of perturbations during training, which improves robustness.

ShareVeo3

ShareVeo3 [71] is a fully synthetic video dataset containing 1,460 videos generated by Google Veo3 [72]. Unlike AVDeepfake-1M, ShareVeo3 videos are entirely fake and lack explicit augmentations from a single source videos. However, we leverage temporal continuity within each video: consecutive frames share semantic content (same scene, same subject) and can be treated as a content group for identity learning. ShareVeo3 serves two purposes: (1) it provides additional fake training data from a different generation method, reducing overfitting to AVDeepfake-1M’s specific artifacts, and (2) it enables evaluation on fully synthetic videos, which differ from partial injection attacks. We generate 63,184 embeddings from 1,460 fully fake videos (no real samples, no partial fakes). During training, we pair ShareVeo3 fakes with real samples from AVDeepfake-1M++ in balanced batches.

Train, Test, and Validation Splits

Sora2 (Out-of-Distribution Evaluation)

To evaluate our model’s ability to generalize to future generative methods, we construct an out-of-distribution test set from OpenAI’s Sora2 [69], widely considered the current state-of-the-art in video generation. We collected the top 150 videos from the Sora2 app’s search

feed, yielding 11,317 embeddings after preprocessing at 0.15-second intervals.

This dataset serves as the critical test of our method’s robustness to novel generation methods. Our hypothesis is that by training the authenticity projection head exclusively on real videos to learn a low-variance manifold of intrinsic real-footage properties (Section 3.3.2), the model will naturally reject Sora2 videos as anomalies because they deviate from the shared statistical regularities that define authentic video data. Crucially, the model has never seen examples from this generation method during training, making this a pure test of generalization to unseen deepfake techniques.

To characterize the distribution of our Sora2 test set, we manually annotated each video across multiple dimensions. Table 3.1 summarizes the metadata distribution.

Table 3.1: Sora2 test set metadata distribution across 150 videos.

Category	Attribute	Count (%)
Human Presence	Contains humans	115 (76.7%)
	Visible faces	110 (73.3%)
Number of People	0 people	39 (26.0%)
	1 person	75 (50.0%)
	2 people	17 (11.3%)
	3 people	4 (2.7%)
	4 people	4 (2.7%)
	5+ people	19 (12.7%)
Speech	Talking subjects	106 (70.7%)
Camera Shot	Close-up	34 (22.7%)
	Medium	87 (58.0%)
	Wide	29 (19.3%)
Camera Motion	Static	41 (27.3%)
	Slight movement	71 (47.3%)
	Heavy motion	38 (25.3%)
Lighting	Good lighting	119 (79.3%)
	Low-light	31 (20.7%)
Content Realism	Natural/Photorealistic	108 (72.0%)
	Stylized	19 (12.7%)
	Surreal	23 (15.3%)

This metadata distribution reveals that our Sora2 test set predominantly contains human-centric, naturalistic content with visible faces and speech, characteristics aligned with the most critical deepfake detection scenarios. However, the substantial inclusion of non-human content (12.7%), stylized and surreal videos (15.3% combined), and diverse cinematographic conditions (varying camera angles, motion, and lighting) is crucial for validating our approach. Strong performance across this heterogeneous content distribution would indicate that our

method learns general intrinsic properties of authentic video data, rather than face-specific forensic artifacts. This distinction is critical: traditional deepfake detectors perform latent forensics, identifying traces left by specific manipulation algorithms. By contrast, our approach performs a deeper *latent analysis*, characterizing the shared structure of real videos across arbitrary content.

Importantly, Sora2 videos contain no real samples or partial injections—all 11,517 embeddings are fully synthetic. During evaluation, we pair these fake embeddings with real samples from AVDeepfake-1M++ to compute detection metrics.

3.4.2 Embedding Selection

3.4.3 Representation Quality Analysis

Clustering-Based Metrics

These metrics test whether embeddings naturally encode the real/fake distinction through their geometric structure, without requiring supervised classification. We apply k-means clustering ($k = 2$) to the z^{auth} embeddings and measure agreement with ground truth labels using three complementary metrics that correct for chance agreement and assess cluster quality.

(i) Adjusted Mutual Information To assess whether our z^{auth} embeddings naturally encode authenticity information in their geometric structure, we compute the Adjusted Mutual Information (AMI) between cluster assignments and ground truth labels [73]. Given ground truth labels U and cluster assignments V from k-means, the mutual information $I(U; V)$ measures the reduction in uncertainty about one partition given knowledge of the other:

$$I(U; V) = \sum_{i=1}^R \sum_{j=1}^C \frac{n_{ij}}{n} \log \left(\frac{n_{ij} \cdot n}{a_i \cdot b_j} \right) \quad (3.10)$$

where $n_{ij} = |U_i \cap V_j|$ is the number of samples in both class i and cluster j , $a_i = \sum_{j=1}^C n_{ij}$ is the size of class i , $b_j = \sum_{i=1}^R n_{ij}$ is the size of cluster j , and n is the total number of samples. However, mutual information increases with the number of clusters even for random partitions. AMI corrects for this by subtracting the expected mutual information under a hypergeometric null model and normalizing by average entropy:

$$\text{AMI}(U, V) = \frac{I(U; V) - \mathbb{E}[I(U^*; V^*)]}{\frac{1}{2}(H(U) + H(V)) - \mathbb{E}[I(U^*; V^*)]} \quad (3.11)$$

where $H(U) = -\sum_{i=1}^R \frac{a_i}{n} \log \left(\frac{a_i}{n} \right)$ denotes the entropy of partition U , and the expectation is taken over random clusterings with the same cluster-size distribution [73]. AMI ranges from 0 (no agreement beyond chance) to 1 (perfect correspondence). High AMI scores indicate that unsupervised clustering of z^{auth} successfully recovers the real/fake distinction, suggesting that authenticity is encoded as an intrinsic geometric property of the embedding space rather than requiring explicit supervision.

(ii) Adjusted Rand Index The Adjusted Rand Index (ARI) provides a complementary perspective by measuring pairwise agreement between clustering and ground truth [74]. Rather than considering information-theoretic quantities, ARI counts the number of sample pairs that are either grouped together or separated in both partitions. The Rand Index is defined as:

$$\text{RI}(U, V) = \frac{a + b}{\binom{n}{2}} \quad (3.12)$$

where a is the number of pairs in the same class and same cluster, and b is the number of pairs in different classes and different clusters. However, like mutual information, the Rand Index achieves non-zero values for random clusterings. ARI corrects for this by subtracting the expected index under random partitions and normalizing by the maximum possible value [74]:

$$\text{ARI}(U, V) = \frac{\sum_{i=1}^R \sum_{j=1}^C \binom{n_{ij}}{2} - \left[\sum_{i=1}^R \binom{a_i}{2} \right] \left[\sum_{j=1}^C \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_{i=1}^R \binom{a_i}{2} + \sum_{j=1}^C \binom{b_j}{2} \right] - \left[\sum_{i=1}^R \binom{a_i}{2} \right] \left[\sum_{j=1}^C \binom{b_j}{2} \right] / \binom{n}{2}} \quad (3.13)$$

where $\binom{x}{2} = \frac{x(x-1)}{2}$ counts the number of unordered pairs from x items. ARI ranges from -1 to 1 , with values near 1 indicating strong agreement, 0 indicating random clustering, and negative values indicating systematic disagreement (though rare in practice). By focusing on pairwise decisions rather than information content, ARI offers robustness to cluster size imbalances and provides interpretable validation of clustering quality.

(iii) Silhouette Coefficient The silhouette coefficient measures how well-separated and compact clusters are by comparing within-cluster cohesion to between-cluster separation [75]. For each sample i , we compute:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (3.14)$$

where $a(i)$ is the mean distance from sample i to all other samples in the same cluster, and $b(i)$ is the mean distance from sample i to all samples in the nearest other cluster. The silhouette coefficient ranges from -1 to 1 , with values near 1 indicating that sample i is well-matched to its cluster and poorly matched to neighboring clusters, values near 0 suggesting that the sample lies on the boundary between clusters, and negative values indicating potential misclassification. The overall silhouette score averages $s(i)$ across all samples, providing a global measure of clustering quality. We compute silhouette scores using both ground truth labels (primary metric, testing embedding space separability) and cluster assignments (validation metric, testing k-means convergence quality). We use cosine distance rather than Euclidean distance to account for the angular relationships in normalized embedding spaces. High silhouette scores indicate that real and fake samples form tight, well-separated groups in z^{auth} , directly validating the success of our variance minimization objective (Equation 3.6).

Distribution-Based Metrics

(i) KL Divergence

(ii) Jensen-Shannon Distance

(iii) Wasserstein Distance

Separation Metrics

(i) Mean Cosine Similarity

(ii) Distance to Real Manifold

Local Content-Group Analysis

(i) Intra-Group Cosine Similarity

(ii) Intra-Group Variance

(iii) Local Orthogonality Coefficient

3.4.4 Detection Performance

In-Distribution Results

Out-of-Distribution Generalization

Temporal Localization (AVDeepfake-1M only)

3.5 Results

3.6 Discussion

Chapter 4

Conclusion

4.1 Summary of Contributions

4.2 Methodological Insights

4.3 Broader Impact

4.4 Future Directions

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